Azam Shahbaz

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Education

Georgia Institute of Technology *Atlanta, GA*
Bachelor of Science in Computer Engineering *May 2028*

Threads: Computing Hardware & Emerging Architectures, Systems and Architecture

Experience

Flock Safety (YC S17) Atlanta, GA
Device Software Engineer Intern *May 2026 – August 2026*

- Developed Linux kernel drivers in C for Qualcomm embedded platforms, exposing **15+** battery telemetry attributes and **5** protection registers to userspace through sysfs and IOCTL interfaces.
- Built an interrupt-driven GPIO driver for **2** PIR motion sensors with character device support and an IOCTL interface for userspace control.
- Designed a C++ battery heater control daemon with a **10-command** IOCTL API and a **3-state** thermal state machine managing regulator and PWM behavior.

Voicu Research Lab at Kennesaw State University Marietta, GA
Undergraduate Researcher *September 2025 – May 2026*

- Contributed to AGICore, a real-time robotics control pipeline, developing **8+** ROS 2 C++ behavior trees to coordinate actuator commands, sensor feedback, and task execution across **3** concurrently running processes.
- Integrated and calibrated **3** sensor modalities (cameras, microphones, IMUs) with ROS 2, achieving sub-**20ms** time-synchronization across sensor streams for real-time edge deployment.
- Profiled hardware-in-the-loop pipelines across **3+** test configurations, evaluating timing, power, and sensor reliability while reducing end-to-end control latency by **25%**.

Intelligent Biomechanics Laboratory at Kennesaw State University Marietta, GA
Undergraduate Researcher *September 2025 – May 2026*

- Implemented and compared **4** segmentation models in Python (including a U-Net) to track organ lumens in de-identified clinical video.
- Processed and organized **13,000+** frames with classical computer vision (thresholding, morphology, contour analysis) in Git-versioned, reproducible Jupyter pipelines.
- Built overlays and visualizations for Emory clinicians, improving procedural analysis speed by up to **50%**.

Projects

Persistent Key-Value Store | *C++20, Linux, CMake* Febuary 2026

- Built a Bitcask-style **log-structured key-value store** with a write-ahead log, CRC32 checksums, and crash recovery; background compaction cuts disk usage by **60%+**.
- Implemented a **multithreaded epoll TCP server** (GET/PUT/DEL) for **64** concurrent clients, reaching **150K+** reads/sec and **40K+** durable writes/sec.
- Verified durability with **1,000** randomized `kill -9` crash-injection runs and **100+** unit tests; zero lost writes.

Thread Pool and Custom Allocator | *C++20, Linux, Valgrind* June 2026

- Implemented a **work-stealing thread pool** with Chase-Lev deques and futures, with a **6**× speedup on **8** threads.
- Wrote a **thread-caching allocator** with size-class free lists, **3**–**5**× faster than `malloc` on small multithreaded allocations.
- Validated with **80+** tests and **1,000+** stress runs under TSan, ASan, and Valgrind; zero data races or leaks.

Multithreaded Packet Analyzer | *C++20, Linux, CMake, Python* September 2026

- Built a **packet analyzer** parsing Ethernet/VLAN/IPv4/TCP/UDP from pcap files and live `AF_PACKET` sockets.
- Designed a **lock-free multithreaded pipeline** sustaining **2M+** packets/sec, with a sharded flow table tracking **100K+** connections and TCP stream reassembly.
- Verified against **tshark** on **50+** captures and fuzzed parsers with libFuzzer; **70+** unit tests.

Technical Skills

Languages: C++20, C, Python, Java, SystemVerilog, RISC-V Assembly, MATLAB, R, LaTeX
Systems: Linux, Kernel Drivers, sysfs, IOCTL, Multithreading, Device Trees, GPIO, I2C
Frameworks/Protocols: ROS 2, Ethernet, IPv4, UDP, CAN, SPI, UART